

DAX functions in Power BI:

1. Aggregation Functions

Used to perform calculations on data columns.

DAX

- SUM(column) - Adds all values in a column
 - AVERAGE(column) - Returns the average of a column
 - MIN(column) - Returns the smallest value
 - MAX(column) - Returns the largest value
 - COUNT(column) - Counts the number of non-blank values
 - COUNTROWS(table) - Counts the number of rows in a table
 - DISTINCTCOUNT(column) - Counts unique values in a column
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2. Logical Functions

Used for conditional calculations.

DAX

- IF(condition, result_if_true, result_if_false)
 - SWITCH(expression, value1, result1, value2, result2, ..., else_result)
 - AND(condition1, condition2) - Returns TRUE if both conditions are true
 - OR(condition1, condition2) - Returns TRUE if at least one condition is true
 - NOT(condition) - Reverses the condition (TRUE to FALSE, FALSE to TRUE)
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3. Date & Time Functions

Used to work with dates.

DAX

- TODAY() - Returns the current date
 - NOW() - Returns the current date and time
 - YEAR(date) - Extracts the year from a date
 - MONTH(date) - Extracts the month (1-12)
 - DAY(date) - Extracts the day of the month
 - EOMONTH(start_date, months) - Returns the last day of the month
 - DATEDIFF(start_date, end_date, unit) - Returns the difference between two dates in days, months, years, etc.
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4. Text Functions

Used to manipulate text data.

DAX

- CONCATENATE(text1, text2) - Joins two text strings
 - COMBINE = 'Customer'[FirstName] & " " & 'Customer'[LastName]
 - LEFT(text, number_of_chars) - Returns the leftmost characters
 - RIGHT(text, number_of_chars) - Returns the rightmost characters
 - MID(text, start, length) - Extracts a portion of a text string
 - SEARCH(substring, text, start_position) - Finds the position of text in a string
 - SUBSTITUTE(text, old_text, new_text) - Replaces text in a string
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5. Filter Functions

Used to filter data in measures or calculated columns.

DAX

- FILTER(table, condition) - Returns a table filtered by condition
 - ALL(table_or_column) - Ignores filters on a column or table
 - ALLEXCEPT(table, column) - Ignores all filters except the specified column
 - KEEPFILTERS(expression) - Keeps applied filters when modifying a calculation
 - RELATED(table[column]) - Fetches data from a related table
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6. Time Intelligence Functions

Used for analyzing data over time.

DAX

- TOTALYTD(SUM(column), DataTable[Date]) - Year-to-date total
 - TOTALQTD(SUM(column), DataTable[Date]) - Quarter-to-date total
 - TOTALMTD(SUM(column), DataTable[Date]) -- Month-to-date total
 - PREVIOUSMONTH(DataTable[Date]) - Returns the previous month
 - NEXTMONTH(DataTable[Date]) - Returns the next month
 - SAMEPERIODLASTYEAR(DataTable[Date]) - Returns the same period last year
 - DATESBETWEEN(DataTable[Date], StartDate, EndDate) -- Filters dates between two values
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7. Mathematical & Statistical Functions

Used for mathematical calculations.

DAX

- ABS(number) - Returns the absolute value
 - ROUND(number, decimals) - Rounds a number to a given number of decimals
 - CEILING(number, significance) - Rounds up to the nearest multiple
 - FLOOR(number, significance) - Rounds down to the nearest multiple
 - POWER(number, exponent) - Raises a number to a power
 - SQRT(number) - Returns the square root
 - RAND() - Returns a random number between 0 and 1
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8. Rank & Sorting Functions

Used to rank values in a dataset.

DAX

RANKX(table, expression, [value], [order]) - Ranks values in a column

9. Relationship Functions

Used to work with related tables.

DAX

RELATED(Table[Column]) - Returns a value from a related table

RELATEDTABLE(Table) - Returns the entire related table

10. Variables in DAX

Variables improve performance and readability.

DAX

VAR TotalSales = SUM(Sales[Amount])

RETURN TotalSales / COUNT(Sales[CustomerID])

11. Using DAX in Measures vs. Calculated Columns

- **Measures:** Dynamic calculations based on filters. Best for aggregations.
- **Calculated Columns:** Computed at the row level and stored in the table. Best for fixed values.